ABHAY RAJ YADAV

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PROFESSIONAL SUMMARY

Final-year Computer Science and Design student (CGPA 8.85/10, graduating 2027) with a Minor in Artificial Intelligence from IIT Ropar, proficient in Java, Python, and AI/ML with hands-on experience building efficient, reusable, and scalable full-stack software – REST APIs, backend microservices, and LLM/agentic AI systems. Software development intern with experience in real-time data acquisition, HMI development, and role-based access control, plus practical exposure to relational and vector databases, Docker, and CI/CD pipelines. Strong analytical thinking, independent debugging ability, and a collaborative approach to building features from scratch and improving existing modules.

TECHNICAL SKILLS

Core Languages: Java, Python, JavaScript, TypeScript, C++, C, SQL, HTML, CSS
AI / ML & Agentic Systems: LLM Integration (Llama 3.3, Gemini API), LangChain, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), Multi-Agent Workflows, Tool/Function Calling
Backend & APIs: Node.js, Express.js, FastAPI, Flask, Django, REST API Design, Microservices
Frontend: React.js, Next.js
Databases: PostgreSQL, MySQL, MongoDB, Redis, Qdrant (vector DB)
Cloud & Tools: Google Cloud, AWS (Cloud Practitioner certified), Docker, Git/GitHub, CI/CD (GitHub Actions), Linux CLI
Core CS Concepts: Data Structures & Algorithms, OOP, Design Patterns, DBMS, Operating Systems, Computer Networking

EXPERIENCE

Software Development Intern | *Micromatic Grinding Technologies* *Sep 2026 – Mar 2027 | Bengaluru*

- Developed an RS232 serial communication module for Statistical Process Control (SPC) software to acquire and process real-time measurement data from industrial machines and instruments.
- Designed and built a Human-Machine Interface (HMI) for a 26-inch display with an on-screen keyboard and Auto, Manual, and other operating modes for shop-floor use.
- Implemented role-based access control (RBAC) with OEM, Admin, Supervisor, and Operator roles to restrict application features and settings by user privilege.
- Collaborated with the engineering team to develop and improve production software for precision grinding machines.

PROJECTS

Niriksha — Agentic AI Honeypot & Threat Intelligence Platform | *Python, FastAPI, Llama 3.3, Groq API, Docker* *Jan 2026 – Present*

- Designed a multi-agent Python backend coordinating an LLM (Llama 3.3 via Groq API) with tool calls and rule-based escalation logic, handling 15+ live fraud scenarios end-to-end.
- Built a regex-based extraction tool invoked by the agent pipeline, achieving 100% parsing accuracy for UPI IDs, bank accounts, and phishing URLs across edge cases.
- Engineered async REST APIs with error-handling and retry logic under load, sustaining a 100% request success rate within a 30-second SLA.

PDF-Summary AI — Enterprise RAG Document Intelligence Platform | *Node.js, Express, TypeScript, Redis, BullMQ, MongoDB, Qdrant, Gemini API* *2026*

- Built an async ingestion pipeline (parse → chunk → embed → index) over a Qdrant vector store, enabling conversational Q&A over documents with citation-backed, page-referenced answers.
- Implemented hybrid retrieval with reranking and similarity-threshold grounding to reduce hallucination and improve answer relevance for enterprise-scale document sets.
- Designed SSE streaming responses and a scalable microservice architecture for real-time, low-latency Q&A – built as a reusable, production-grade SaaS module.

DemandRadar — Scalable Analytics & Demand Forecasting Platform | *Python, React.js, MongoDB, Flask, REST APIs Jan 2024 – May 2024*

- Built a full-stack analytics platform (React.js frontend, Flask/MongoDB backend) with Python/Flask REST APIs handling 1,000+ concurrent requests at 99% uptime.
- Delivered the platform with a 4-member team using Agile/Scrum methodology, implementing caching to optimize query performance under load.

EDUCATION

Atria Institute of Technology *Sep 2023 – Sep 2027*
B.E. in Computer Science and Design Bengaluru, Karnataka
CGPA: 8.85/10.0 | Coursework: DSA, OOP, Design Patterns, DBMS, OS, Computer Networks, Software Engineering

Indian Institute of Technology, Ropar *Sep 2024 – Sep 2025*
Minor in Artificial Intelligence Punjab, India
Coursework: Neural Networks, LLMs, Computer Vision, NLP, Prompt Engineering, Generative AI

ACHIEVEMENTS & CERTIFICATIONS

- **Top 2% globally** at AI Impact Summit BuildAthon 2026 (international product engineering competition); **2nd Runner-Up** at Inter-College Ideathon
- **Certifications:** AWS Cloud Practitioner, Network Fundamentals, Programming in C, Python Design Patterns, Cybersecurity MOOC (CySecK)
- **Open Source:** 50+ GitHub contributions across Python, data science, and machine learning repositories